REPETITION, NOT LENGTH: ISOLATING THE COUNTING FAILURE IN NEURAL TEXT-TO-SPEECH

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ABSTRACT

Text-to-speech models loop, truncate and lose count on text that repeats a phrase many times. We show that repetition itself is what breaks them, not the length that comes with it. Every repeated sentence in our test set is paired with a control of matched sentence and word count in which no word ever repeats back-to-back. Six models from three architectures render the controls almost perfectly and fail the repeated twins: 94.3% against 18.2% exactly right at $k \geq 6$. The gap survives greedy decoding, repetition-penalty sweeps, four independent speech recognisers and 420 analysis specifications without once reversing sign; a held-out fourth architecture lands within a point of its predicted gap, and one of two non-autoregressive baselines shows the same failure. Varying the period of the text shows the failure grows smoothly with periodicity, half of it surviving when no word is adjacent to itself.

Index Terms— speech synthesis, hallucination, repetition, autoregressive models, evaluation

1. INTRODUCTION

Ask a text-to-speech (TTS) model to repeat a word a dozen times and it will frequently get the count wrong: it stops early, runs on, or locks into a loop. The behaviour is documented as a stability problem [1,2] and usually attributed to exposure bias or missing alignment constraints, with fixes proposed at the level of attention supervision or decoding rules [3–5]; it is the speech analogue of neural text degeneration [6,7].

Such reports confound two variables, because repeating a phrase k times makes the text both longer and repetitive. If long inputs are simply harder, nothing about repetition needs explaining. Our design separates the two: every repeated item is paired with a *length-matched control* of the same carrier and word count, in which the k copies of one word become non-adjacent distinct fillers, drawn from a fixed pool of eight. A length account predicts the two curves coincide. They do not.

We contribute: (i) a controlled manipulation isolating repetition from length, validated on six checkpoints, two non-autoregressive baselines and a held-out architecture; (ii) a purpose-built test set and a judge methodology for repetitive speech; and (iii) a negative mechanistic result: the contraction account of [8] does not describe any decoder we measured.

Table 1. Likelihood of the paired texts under two independent scorers: mean NLL per token (lower = more probable) and the share of 54 pairs whose control is less probable. A panel-internal scorer agrees (supplementary material).

Scorer	NLL/token		ctl. less probable
	rep.	ctl.	(% of pairs)
phi-2	3.69	4.89	87
GPT-2-large	3.36	5.34	100

2. EXPERIMENTAL SETUP

Test set. 180 items, generated deterministically by a released script; the texts are authored, not drawn from a corpus, since every sentence must contain its target word a known number of times. Templates were drafted with the Claude Fable 5 language model[1] and edited by hand, under fixed constraints: targets are common one- or two-syllable words without close homophones, and carriers are plain conversational English. Six carrier templates each fix a prefix, target and suffix ("The dog was *very* … big"); the target is repeated $k \in \{1, 2, 3, 4, 6, 8, 12, 16, 24, 32\}$ times; we call this sweep of k the ladder. Sentence-repetition, tongue-twister and number-phrase item types are built the same way. Every item at $k \geq 2$ has a length-matched control from the same template: the k copies are replaced, in fixed order, by fillers from a per-template pool of eight, so only repetition differs within a pair. The pool cycles above k=8, making those controls period-8; a never-cycled variant quantifies the difference (Sec. 3). Nor are the items improbable strings (Table 1): under two independent language models the *control* is the less probable text in most pairs, so the repeated items are, if anything, the more natural ones. The generation script, test set, transcripts and analysis code are released.[2]

Models. Six checkpoints from three architectures form the panel: Llasa at 1B/3B/8B (Llama backbones over X-codec2 tokens [9]), Qwen3-TTS at 0.6B/1.7B [10], and XTTS-v2 [11]. Outside the panel: VITS [12] and F5-TTS [13] as non-autoregressive baselines, CosyVoice 2 [4] as a fourth architecture held out until the analysis was frozen, and penalty, greedy and sampling ablations of panel members (Tables 3 and 6). Every system renders every item at three seeds under

[1]https://www.anthropic.com/news/claude-fable-5-mythos-5

[2]https://github.com/lab260ru/icassp_antispoofing

each system's shipped decoding defaults. The defaults make Qwen3-TTS the most stochastic family, so its larger checkpoint's near-zero count-error gap (Sec. 3) is not conservative decoding at work.

Judge. The judge, the speech recogniser whose transcript we count, is a CTC model (`wav2vec2-large-960h-lv60-self`; greedy best-path, no language model). The natural choice, Whisper large-v3 [14], is unsuitable here: its decoder is autoregressive with a language-model prior and hallucinates on exactly the audio under study [8]. On concatenated utterances whose true count is known by construction, Whisper recovers a median 0.25 of the count on the 48 trials where it returns anything, and nothing at all in the other 60 of 108; the CTC judge recovers 1.00 (n=108) on the same audio. The full audit and the four-recogniser replication are in the supplementary material.

Metrics. $\hat{c}$ is the number of renditions of the target found in the transcript, counted in order and non-overlapping, skipping unrecognised words (stopping at the first miss would void every later filler). Three quantities are reported. *Relative count error*: $(\hat{c} - k)/k$, negative for undercounts. *Exactly right*: the share of generations with $\hat{c} = k$. *Gap*: control minus repeated exactly-right rate, in percentage points; seen from the repeated side we also call it the deficit. Headline analyses cover the word-repetition ladder and its controls (other item types: supplementary material). Exclusion rules blind to the count (a template the judge cannot transcribe, our token budget, degenerate audio) keep 1558 of 2052 generations; the accounting is in the supplementary material and Table 6 shows the gap with every rule off. Individual analyses run on slightly different populations, so each number carries its own n.

Uncertainty. Every bracketed range $[\cdot, \cdot]$ in this paper is a 95% confidence interval: Wilson for single rates; for panel summaries, a nonparametric bootstrap over checkpoints, the unit our claims generalise over (generations cluster within one). Every headline estimate is computed four ways (per checkpoint, per family, mixed-effects and hierarchical Bayes; Table 2) and under 420 specifications, so no conclusion rests on one analysis choice.

3. RESULTS

3.1. The limit tracks repetition, not length

Figure 1(a) is the central test. Each control carries its twin's carrier and word count, so any account in which length is what hurts (attention span, exposure bias, duration budgets) predicts the two curves coincide. They do not: from k=6 the controls hold at 94.3% exactly right while repeated items fall from 40.5% at k=6 to 6.8% at k=32, 18.2% pooled (Table 3). Below k=6 the two curves do not separate.

The miscounts are large and biased toward undercounting. Repeated items run a median count error of -12.5% per checkpoint against 0.0% [0.0, 0.0] for controls (n=525), at every k up to 32, in five of six checkpoints; the sixth ties at zero on this metric yet still loses 60.0 points of exact rate. Before exclusions, at $k \geq 8$ (n=540), 51.3% truncate or undercount and 36.3% loop or overcount.

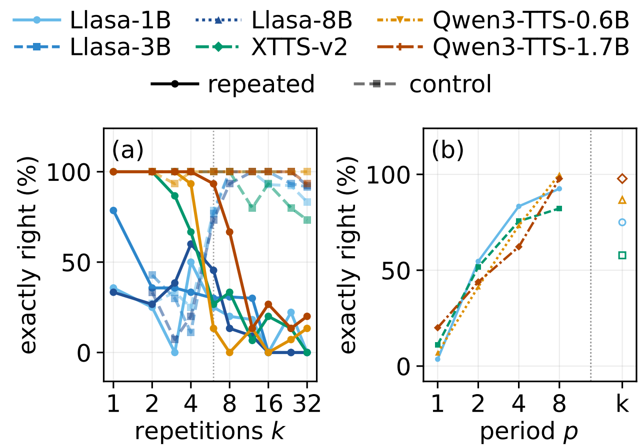

Fig. 1. The repeated–control dissociation, in the paper's headline metric. (a) Exactly-right renderings against k: from k=6, length-matched controls (dashed) stay near-perfect while repeated text (solid) collapses; below k=6 transcription noise affects both conditions. (b) On the four checkpoints measured, at fixed carrier, word count and requested count, exact renderings rise smoothly with period p; hollow markers: the all-distinct p=k items.

Table 2 gives the headline estimates under every clustering treatment. The exact-rate gap is positive in all 6 of 6 checkpoints and all 3 families, and all six interval constructions of the mixed-effects model exclude zero.

The statistical model also predicts out of sample: partial pooling of the three architectures puts the gap for an unseen architecture at 65.3 [5.9, 93.3] points; the widest prior widens that interval to include zero, so we tested the prediction. CosyVoice 2, the design most likely to be immune thanks to its alignment supervision, its numbers predicted before it generated a waveform, lands within a point of the prediction, at 64.4 (Tables 2 and 3).

The $k \geq 6$ threshold was chosen after seeing where the curves diverge, so we also quote the unconditioned whole-ladder gap (Table 2) and vary the threshold along with every other analysis choice: across 420 specifications over six axes (exclusions, threshold, control variant, seeds, aggregation, outcome definition) the gap spans 34.4–86.4 points, median 70.5, and 0 of 420 reverse sign (Table 6).

Above k=8 the cycled control is itself period-8, and the headline gaps above use that cycled control. Using never-cycled fillers throughout shrinks the per-family $k \geq 6$ gap from 75.4 to 67.6 points and the per-checkpoint one from 76.7 to 69.6. Most of the difference is vocabulary difficulty, not periodicity: 1.7 points survive one unit of counting tolerance, and none of it shows on median error. Re-generated at matched k (12–32) with fully aperiodic controls, the rates are 10.0% against 83.1% (n=349): 73.1 points with no cycling anywhere.

Extension to $k \leq 128$ (exploratory). Within $k \leq 32$ the deficit is proportional in k, not saturating (residual 2.6 against 485.3 for a saturating fit), so no collapse point lies inside the main ladder. Extending 4 checkpoints to $k \leq 128$ shows the

Table 2. Headline estimates with 95% intervals, in exactly-right points (the count-error row: points of median relative error). The last two rows are the pre-registered out-of-panel prediction and its test.

Quantity	est.	95% interval
Exact-rate gap, per checkpoint (n=6)	76.7	[68.4, 84.4]
Exact-rate gap, per family (n=3)	75.4	[71.1, 78.9]
Exact-rate gap, mixed effects (widest of six)	76.4	[63.2, 89.2]
Count-error gap, per checkpoint	10.4	[5.6, 13.9]
Exact-rate gap, whole ladder (n=1559)	46.8	—
Predicted gap, unseen architecture	65.3	[5.9, 93.3]
Observed, CosyVoice 2 [4] (n=90 per arm)	64.4	—

Table 3. Every system tested, $k \geq 6$, CTC judge: median $(\hat{c} - k)/k$ and exactly-right rate, repeated (rep.) against control (ctl.). Systems in the lower block are outside the panel; their count errors are in the supplementary material.

Model	size	count err.		exact (%)	
		rep.	ctl.	rep.	ctl.
Llasa-1B [9]	1B	−12.5	0.0	13.3	91.5
Llasa-3B [9]	3B	−8.3	0.0	16.9	94.1
Llasa-8B [9]	8B	−16.7	0.0	11.8	93.2
XTTS-v2 [11]	0.4B	−12.5	0.0	16.7	87.8
Qwen3-TTS-0.6B [10]	0.6B	−12.5	0.0	7.9	100.0
Qwen3-TTS-1.7B [10]	1.7B	0.0	0.0	38.9	98.9
Panel, pooled	6 ckpts.	−8.3	0.0	18.2	94.3
CosyVoice 2 [4]	held-out	—	—	28.9	93.3
F5-TTS [13]	non-AR	—	—	25.6	85.6
VITS [12]	non-AR	—	—	65.6	58.9

two conditions diverging in absolute terms (Table 4): the median rendered count of repeated items *falls* as the requested count rises, while controls keep climbing. The fitted saturation scale drops 4.2-fold, but the factor depends on the fitted form (3.1–5.4 across three saturating forms) and is not unanimous (in Qwen3-TTS-1.7B it moves the other way), so this cannot tell a hard limit from ordinary saturation.

3.2. The deficit is graded in the period

So far, repetition and periodicity vary together; a second ladder holds carrier, word count and requested count fixed and varies only the period p. The deficit is graded, not a step: exact rates rise 10.4, 47.9, 73.6, 93.1% at p=1, 2, 4, 8, monotone in all four checkpoints measured (Table 5; Fig. 1b).

The decisive step is p=2: no token is ever adjacent to itself, yet half the deficit remains (0.547 on the strictest of five scoring rules, 0.69–0.86 on the others). Nor is it a power-of-two artifact: at k=24 the odd step p=3 (62.3%) and non-power p=6 (90.0%) land between their neighbours, as pre-registered. A comma, full stop or "and" between repetitions recovers none of it (n=465): the copies are segmented, just miscounted.

Whether the *ordering* of a period-p text or its restricted *vocabulary* carries the effect is unresolved. Shuffling each item's own words recovers almost none of the gap pooled (0.6%), but that pooled number averages over a split: two checkpoints gain

Table 4. The $k \leq 128$ extension, 4 checkpoints: median *rendered* count against the requested count. Repeated renderings regress while controls advance.

Requested count	repeated	control
k=48	25.0	46.0
k=64	25.0	59.5
k=96	22.0	64.0
k=128	17.0	66.5

Table 5. The period ladder: exactly-right rate (%) against the period p of the text at fixed carrier, word count and requested count, pooled over the four checkpoints measured. The second row re-measures at k=24 with the odd and non-power steps added.

	p=1	p=2	p=3	p=4	p=6	p=8
Ladder scan (n=1539)	10.4	47.9	—	73.6	—	93.1
At k=24 (n=1202)	8.5	51.2	62.3	75.2	90.0	92.9

17.8 points, one is null, and XTTS-v2 loses 30.0. That split is why the title claims repetition rather than periodicity.

3.3. Not the decoding rule or our analysis choices

Table 6 varies the decoding rule and the analysis. The families ship penalties differing by an order of magnitude; swept over 4-fold (XTTS-v2) and 3-fold (Qwen3-TTS-0.6B) ranges, the gap stays put (15.6–21.1% and 7.8–16.7% exact). The three Llasa checkpoints apply no penalty at all and show it at full size; with the penalty disabled entirely, Qwen3-TTS-0.6B renders 10.0% exactly against 100.0%.

Greedy decoding and repetition-aware sampling leave the gap in place. Nor do our exclusion rules create the gap: with every rule off it is 61.7 points. The one rule that falls unevenly, the token budget (hit by 10.6% of repeated against 5.6% of control generations), can only cut counts short; keeping those rows would widen the gap, so excluding them is conservative. Re-scored with four independent recognisers it stays at 65.1–76.7 points, positive everywhere; the per-judge analysis is in the supplementary material.

3.4. Non-autoregressive baselines

VITS shows no dissociation on the same strings through the same judge: −6.7 points against the panel's 76.1 (Table 3). F5-TTS, flow-matching rather than autoregressive, shows a 60.0-point gap, so the failure is not specific to autoregressive decoding.

What plausibly separates the two is where the count must live: VITS predicts a duration per token, F5-TTS one total. Handed the oracle total duration, F5-TTS improves markedly below k=12 and not at all at or above it (Table 7): given the right length, the model still cannot place thirty-two repetitions inside it.

3.5. Repetition slows the growth of the state

The failure is also visible in the decoder's state space. On a 136-item subset we record per-step hidden states at deep

Table 6. The gap under decoding and analysis variations. RAS is repetition-aware sampling [5].

	gap (pts)
Panel, $k \geq 6$	76.1
Exclusions off (n=1620)	61.7
No penalty (Llasa ×3)	78.9
Greedy (Qwen3-TTS-0.6B)	72.2
Sampled (Qwen3-TTS-0.6B)	75.0
RAS on (Qwen3-TTS-0.6B)	94.3
RAS off (Qwen3-TTS-0.6B)	92.1
Across four recognisers (range)	65.1–76.7
Across 420 specifications (range)	34.4–86.4

Table 7. F5-TTS handed the oracle total duration: exactly-right rate (%).

	free	oracle duration
Below k=12 (n=30)	43.3	76.7
At or above k=12 (n=60)	16.7	16.7

probe layers and compute the trajectory's effective rank $N_{\text{eff}} = \exp(-\sum_i p_i \log p_i)$, p_i the normalised singular values: the number of distinguishable states visited. The statistic is the *capacity gain* $dN_{\text{eff}}/d \log k$, fitted separately on repeated items and on their length-matched controls.

Repeated text grows new states at 0.46 of its control's rate, median across the panel, with the two intervals disjoint in 5 of 6 checkpoints (Table 8). The gain stays bounded away from zero in 5 of 6, so this is a slowing, not a halt. At checkpoint level the gap in gains is 24.7 $[13.7, 36.5]$ states per unit $\log k$; by family, 20.1 $[10.7, 38.5]$. The ceiling columns of Table 8 agree.

4. DISCUSSION

A mechanism tried, and rejected. The contraction account of decoder hallucination [8] attributes looping to attractor dynamics: repeated conditioning drives the decoder state toward a fixed point, and once states have contracted together no readout can recover the count. It motivated this study; it fails here.

We estimated the contraction factor q: the largest singular value of the map from the decoder state at one repetition boundary to the next. $q < 1$ means states drift together; $q > 1$, that they separate. Across five checkpoints q runs 21.0–347.7: expansive in every item, on repeated and control text alike. It grows with scale inside Llasa, and the one checkpoint with zero median count error sits at q=24.7. Nor is attention near-uniform over the k copies (logit spread 1.23 nats, a 3.4-fold ratio), and flatter attention predicts *better* counting (r=0.59), the opposite of the account's prediction.

What fails is not storage. The requested count stays decodable from late hidden states while the rendering fails (ridge probe, median R^2=0.48), as in text models [15]. Past the ladder (k=48–128) that decodability thins (Table 9): one checkpoint fails to beat a constant predictor, a second ties it (0.99), and on the two Qwen checkpoints, where the probe still wins, a probe given only the audio length does as well, so the surviving signal need not encode the count. Deciding between state and readout needs an intervention at fixed state; our rank-1 patches fail their own positive control, so the question stays open.

Table 8. State capacity per checkpoint: growth of distinguishable states $dN_{\text{eff}}/d \log k$ with 95% intervals, and the saturating-fit ceiling of N_{eff}, repeated against control.

Model	capacity gain		ceiling	
	rep.	ctl.	rep.	ctl.
Llasa-1B [9]	18.2 [9.8,25.7]	56.7 [50.9,63.5]	181	247
Llasa-3B [9]	12.4 [−1.8,23.3]	58.9 [52.8,68.0]	178	249
Llasa-8B [9]	15.0 [11.0,20.3]	45.5 [37.2,53.1]	188	228
XTTS-v2 [11]	24.0 [16.9,31.1]	34.8 [29.1,40.7]	81	105
Qwen3-TTS-0.6B [10]	20.9 [16.3,25.9]	34.9 [30.6,39.6]	81	115
Qwen3-TTS-1.7B [10]	33.4 [29.6,37.2]	41.5 [37.7,45.8]	107	133

Table 9. Probe MAE past the ladder (k=48–128) as a fraction of a constant predictor's; below 1 the count is still decodable. The control column runs the same probe on control items.

Checkpoint	repeated	control
Llasa-1B [9]	1.01	0.86
Llasa-3B [9]	0.99	0.83
Llasa-8B [9]	0.86	0.59
Qwen3-TTS-0.6B [10]	0.74	0.95
Qwen3-TTS-1.7B [10]	0.90	0.89

Related work. Closest is representational collapse [16]; Sec. 3.5 measures it under a length-matched control and finds a slowing, not a fixed point. Counting and state-tracking studies [17–22] delimit what transformers can count, but those limits are model properties, invariant to the period of the input; our effect moves with it.

Limits. Repetition is not separated from training-set frequency, though Table 1 points the wrong way for the obvious form of that confound. The judge is validated on audio, not listeners; the panel, six checkpoints in three families, is as wide as compute allowed.

5. CONCLUSION

With length held fixed, repetition alone breaks every autoregressive text-to-speech system we tested: six checkpoints from three architectures render controls exactly in 94.3% and repeated text in 18.2%: a 76.1-point gap, per checkpoint 76.7 $[68.4, 84.4]$, that nothing we varied reverses (34.4–86.4 across 420 specifications). The gap is graded in the text's period, half remaining with no adjacent repeats. It extends to a held-out architecture (64.4 points, as predicted) and one of two parallel decoders: not autoregression itself. Repeated text grows distinguishable states at 0.46 of the control's rate, and the attractor mechanism fails its own measurement (q never below 1). Why the readout fails is the open question.

COMPLIANCE WITH ETHICAL STANDARDS

This is a computational study of publicly released text-to-speech checkpoints, evaluated on synthetic text written by the authors. It involved no human subjects, no animal subjects and no personal data, and no ethical approval was required. The authors declare no competing financial interests.

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